

AIT362 Programming in R

Previous Question Paper Analysis

Module 1: R Basics and Control Structures

PART A Questions (3 Marks Each)

1. Explain data frames in R with an example. (2024 Dec)
2. Explain vector in R with an example. (2024 Dec)
3. Write an R program to check whether a number is odd or even. (2024 May)
4. Explain matrix in R with an example. (2024 May)

PART B Questions (14 Marks Each)

1. Write an R program to check whether a number is prime number or not. (8 marks) (2024 Dec)
2. Explain data structures in R program. (6 marks) (2024 Dec)
3. Write a R program to check whether a number is armstrong or not. (8 marks) (2024 Dec)
4. Explain with examples for loop, while loop and controlling loops in R. (6 marks) (2024 Dec)
5. Write an R program to find the factorial of a number. (8 marks) (2024 May)
6. Illustrate different vector operations in R with examples. (6 marks) (2024 May)
7. Explain with examples if, if-else and switch statements in R. (6 marks) (2024 May)
8. Write an R program to extract 3rd and 5th rows with 1st and 3rd columns from a data frame. (8 marks) (2024 May)

Key Focus Areas for Module 1

- Data structures in R (vectors, matrices, data frames, lists)
- Control structures (if-else, switch statements, loops)
- Basic R programming (numeric operations, logical testing)
- Data frame manipulation and access
- Vector operations and manipulation

Module 2: Data Manipulation and Import/Export

PART A Questions (3 Marks Each)

1. Explain aggregate function in R. (2024 Dec)
2. Illustrate different ways to access a subset of a dataset. (2024 Dec)
3. List different methods used in binning data. (2024 May)

4. Explain why data cleaning is considered an important task in data analysis. (2024 May)

PART B Questions (14 Marks Each)

1. Explain how data is exported from database in R programming. (7 marks) (2024 Dec)
2. Write an R program to export the following data to a csv file. (7 marks) (2024 Dec)
3. Given a file "auto.csv" of automobile data with the fields index, company, body-style, wheel-base, length, engine-type, num-of-cylinders, horsepower, average-mileage, and price. Write an R program to print total cars of all companies and find the average mileage of all companies. (7 marks) (2024 Dec)
4. Explain different methods used for combining data sets in R. (7 marks) (2024 Dec)
5. Write an R program to create a data frame using two vectors and display the duplicate elements and unique rows in the data frame. (7 marks) (2024 May)
6. Illustrate transformation functions in R. (7 marks) (2024 May)
7. Explain how data is imported from external files with an R program. (7 marks) (2024 May)
8. Write an R program to export the following data to a CSV file. (7 marks) (2024 May)

Key Focus Areas for Module 2

- Data import and export techniques (CSV, databases)
- Data subsetting and aggregation
- Data cleaning and transformation
- Working with data frames
- Combining and merging datasets
- Data binning methods

Module 3: Statistical Analysis

PART A Questions (3 Marks Each)

1. Define probability distribution in R. (2024 Dec)
2. Write an R program to compute the covariance between two vectors using pearson method. (2024 Dec)
3. Define normal distribution in R. (2024 May)
4. Explain the t-test. (2024 May)

PART B Questions (14 Marks Each)

1. Explain different statistical tests performed in continuous data. (7 marks) (2024 Dec)
2. Explain data analysis in R. (7 marks) (2024 Dec)

3. Explain different non-parametric tests in R. (7 marks) (2024 Dec)
4. Explain t-test and ANOVA in R. (7 marks) (2024 Dec)
5. Explain how the statistical test is performed using R functions. (7 marks) (2024 May)
6. Explain the common distribution type arguments used in R functions for statistical analysis. (7 marks) (2024 May)
7. Explain different summary statistics functions in R. (7 marks) (2024 May)
8. Explain different parametric tests in R. (7 marks) (2024 May)

Key Focus Areas for Module 3

- Probability and normal distributions
- Statistical tests (parametric and non-parametric)
- T-test and ANOVA implementation
- Covariance calculations
- Summary statistics functions
- Data analysis techniques

Module 4: Data Visualization

PART A Questions (3 Marks Each)

1. Explain the function used to plot histogram with an R program. (2024 Dec)
2. Explain box plots with the help of an R program. (2024 Dec)
3. Explain the function used to plot scatter plots with an R program. (2024 May)
4. Explain how to plot a pie chart in R. (2024 May)

PART B Questions (14 Marks Each)

1. Explain graphics devices used in data visualization. (7 marks) (2024 Dec)
2. Compare and contrast the ggplot and lattice functions. (7 marks) (2024 Dec)
3. Differentiate bar chart and histogram in data visualization in R. (7 marks) (2024 Dec)
4. Write a R program and explain how to create a Bar Chart. (7 marks) (2024 Dec)
5. Explain ggplot() with an example. (8 marks) (2024 May)
6. Write R program to create pie chart for the following data. (6 marks) (2024 May)
7. Explain lattice function in data visualization. (8 marks) (2024 May)
8. Illustrate customizing charts in data visualization. (6 marks) (2024 May)

Key Focus Areas for Module 4

- Basic plot types (histogram, box plot, scatter plot, pie chart, bar chart)

- Visualization libraries (ggplot2, lattice)
- Graphics devices and export options
- Chart customization techniques
- Comparison between different plotting methods

Module 5: Regression Analysis

PART A Questions (3 Marks Each)

1. Describe the unusual observations in the regression models. (2024 Dec)
2. Explain poisson regression in R. (2024 Dec)
3. Explain Nonlinear least squares. (2024 May)
4. State the applications of regression models. (2024 May)

PART B Questions (14 Marks Each)

1. Distinguish between simple and multiple regression analysis and explain its applications when working with numerical and categorical data. (7 marks) (2024 Dec)
2. With the help of an example write the steps to establish a regression. (7 marks) (2024 Dec)
3. List the different types of regression models. (7 marks) (2024 Dec)
4. Given two vectors, write an R program to predict the weight of new person using regression model. (7 marks) (2024 Dec)
5. Explain corrective measures of unusual observations in regression modelling. (8 marks) (2024 May)
6. Compare and contrast logistic regression with poisson regression. (6 marks) (2024 May)
7. Explain linear and multiple regression. (7 marks) (2024 May)
8. Illustrate model fitting in simple linear model. (7 marks) (2024 May)

Key Focus Areas for Module 5

- Linear and multiple regression
- Specialized regression types (poisson, logistic)
- Model fitting and prediction
- Handling unusual observations
- Applications of regression models
- Nonlinear least squares

Summary of Repeated Questions

Repeated Question Patterns

1. **Data structures explanation** (vectors, matrices, data frames)

- 2. **Control flow implementation** (loops, conditionals)
- 3. **Data import/export operations** - particularly with CSV files
- 4. **Statistical tests explanation** (parametric vs. non-parametric)
- 5. **Visualization libraries comparison** (ggplot vs. lattice)
- 6. **Regression types and applications**

Most Frequently Tested Topics

- 1. **Creating and manipulating basic R data structures**
- 2. **Import/export operations with external data**
- 3. **Statistical analysis functions and methods**
- 4. **Data visualization techniques**
- 5. **Regression analysis and modeling**

Exam Weightage Analysis

Module	PART A Questions	PART B Questions	Overall Percentage
Module 1: R Basics and Control Structures	4	8	20%
Module 2: Data Manipulation and Import/Export	4	8	20%
Module 3: Statistical Analysis	4	8	20%
Module 4: Data Visualization	4	8	20%
Module 5: Regression Analysis	4	8	20%

The exam maintains a perfect balance across all modules, allocating equal weight to each area.

Question Type Analysis

Question Type	Count	Percentage
Theoretical explanation	26	65%
Programming implementation	14	35%

Study Strategy Recommendations

Based on the analysis of previous question papers (May 2024 and December 2024), the following study approach is recommended:

- 1. **Practice basic R programming tasks:** Focus on implementing fundamental algorithms like prime number checking, factorial calculation, and Armstrong number verification.
- 2. **Master R data structures:** Become proficient with creating and manipulating vectors, matrices, data frames, and lists, as these form the foundation of most R operations.

3. **Learn data import/export thoroughly:** Practice reading from and writing to CSV files, as this appears frequently in both exams.
4. **Understand statistical functions:** Be familiar with both the theoretical concepts and practical implementation of various statistical tests, distributions, and analysis methods.
5. **Develop visualization skills:** Practice creating various plot types using both base R graphics and advanced libraries like ggplot2 and lattice.
6. **Study regression techniques:** Understand the theory behind different regression models and be able to implement them in R code.
7. **Balance theory and practice:** While theoretical questions constitute the majority (65%), programming implementation questions carry significant weight (35%), so be prepared to write functional R code.

Conclusion

The AIT362 Programming in R course examination has a well-structured format with equal emphasis on all five modules. The questions test both theoretical understanding and practical programming skills, with a slight lean toward theoretical knowledge. Students should ensure they can not only explain R concepts but also implement them in working code. The pattern of questions suggests that mastering the fundamentals of each module, particularly the repeated topics identified above, will provide a strong foundation for success in the examination.